James Inah

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Education

University of Minnesota — Enrolling Fall 2026
B.S. Computer Science, College of Science and Engineering — *Minneapolis, MN*

Century College — Aug. 2025 – Present
Transfer Coursework in Computer Science, GPA: 3.84/4.0 — *White Bear Lake, MN*

- Coursework: Discrete Structures, Database Management Systems, OOP, C++, Python

Carnegie Mellon University (Open Learning) — Jan. 2026 – Present
Introduction to Modern AI (10-202) — Prof. Zico Kolter — *Remote*

- Implemented a minimal LLM from scratch: custom tokenization, self-attention, and fine-tuning in PyTorch

Projects

Park&Go | *Python, FastAPI, React, TypeScript, PostgreSQL, MapLibre GL, Docker* — Dec. 2025 – Present

- Sustained **18.7 req/sec** under 50 concurrent users with **0 failures across 1,109 requests** and 150ms median latency, verified via Locust load testing; built a Redis caching layer with 5-min TTL across all read-heavy endpoints, dropping cached response times to sub-50ms in production
- Engineered a **6-factor weighted scoring algorithm** (cost, travel time, preferences, campus proximity, verification status, event adjacency) returning ranked parking recommendations in **under 300ms** with no cache; integrated Haversine distance computation and multi-mode travel time estimation (drive/walk) across 3 UMN campuses
- Built **13 RESTful API endpoints** across 6 routers with async FastAPI and SQLAlchemy `AsyncSession` — zero blocking DB calls; added Google OAuth SSO with JWT session management and rate limiting on submission endpoints via SlowAPI
- Automated iCal ingestion from UMN's live campus calendar into PostgreSQL via APScheduler, enabling event-aware recommendations; instrumented Prometheus metrics on all routes plus 4 custom business counters for production observability
- Shipped **123 passing tests** (54 Python pytest + 69 frontend Vitest) spanning unit, integration, and E2E suites, covering geolocation mocks, nav store state transitions, and ETA population across a **10,000+ line codebase**

Text Summarization, Seq2Seq Model | *Python, PyTorch, NLTK, NumPy, scikit-learn* — Mar. 2026

- Built a Seq2Seq model with Luong attention: 3-layer stacked LSTM encoder + 1-layer LSTM decoder trained on 500k+ Amazon food reviews
- Achieved 90% summarization accuracy; condenses full reviews to single sentences with minimal loss
- Custom preprocessing pipeline: HTML stripping, contraction expansion, Lancaster stemming, mode-based length selection, and from-scratch word2idx vectorization

Technical Skills

Languages: Python, Java, JavaScript/TypeScript, SQL, C++
Frameworks & Tools: FastAPI, React, PyTorch, SQLAlchemy, Docker, MapLibre GL, Tailwind CSS, Git
Databases: PostgreSQL, MySQL
Certification: Supervised Machine Learning: Regression & Classification (DeepLearning.AI)

Student Involvement

ColorStack — Jan. 2026 – Present
Member — *Remote*

- Engaged with a national community of 10,000+ Black and Latinx CS students focused on increasing representation in tech
- Collaborated with peers on resume reviews, interview prep, and internship referral pipelines